

Fine-Tuning Pretrained Backbones for Open-Vocabulary Remote Sensing Segmentation

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Abstract

Open-vocabulary remote sensing image semantic segmentation (RSISS) enables models to recognize arbitrary semantic categories at inference time, but existing approaches rely heavily on domain-specific training and struggle to scale efficiently. In this work, we revisit the GSNet framework and study the impact of scaling pretrained generalist and specialist backbones for open-vocabulary RSISS. We replace the original DINOv1 specialist trained from scratch with larger pretrained DINOv3 model and scale the generalist backbone from CLIP ViT-B/16 to ViT-L/14, also adopting selective fine-tuning strategies to balance performance and computational efficiency.

Experiments on the LandDiscover50K dataset and multiple external benchmarks show that selectively fine-tuning pretrained DINOv3 specialists yields stable and competitive performance while substantially reducing training cost. Scaling the generalist backbone further improves open-vocabulary generalization, achieving the strongest results in this study on FAST, FLAIR, and FloodNet. However, performance on ultra-high-resolution datasets such as ISPRS Potsdam remains lower than that of domain-specific specialists trained from scratch, highlighting a trade-off between semantic generalization and spatial precision. Overall, our findings demonstrate that scaling and fine-tuning pretrained foundation models is a practical alternative to specialist pretraining for open-vocabulary RSISS, while also exposing clear limitations that motivate future work. [GitHub](#)

1 Introduction

Remote sensing image semantic segmentation (RSISS) plays an important role in understanding Earth observation data and supports a wide range of real world applications, such as land cover monitoring, environmental protection, and disaster response. While recent deep learning approaches have significantly improved RSISS performance, most existing methods are limited to a predefined set of semantic classes and show reduced generalization to unseen categories or new domains. This limitation becomes particularly problematic in large scale and rapidly changing scenarios, where exhaustive data annotation and retraining are impractical.

Open vocabulary learning offers a promising direction to address these challenges by enabling models to segment arbitrary semantic classes specified at inference time. However, directly applying open vocabulary segmentation methods developed for natural images to the remote sensing domain often leads to suboptimal performance, due to the unique characteristics of remote sensing imagery, such as high spatial resolution, domain shifts, and semantic ambiguity. Recent research has shown that combining generalist vision language representations with remote sensing specific domain knowledge can effectively balance semantic generalization and spatial precision.

In this work, we study the open vocabulary remote sensing image semantic segmentation problem through the replication and extension of a representative state of the art framework on the LandDiscover50K dataset. Our objective is to analyze the effectiveness of integrating generalist and specialist representations in this setting, and to explore potential directions for improving robustness and adaptability while preserving open vocabulary capability. This study aims to provide insights into the strengths and limitations of current approaches and to contribute toward more flexible and generalizable RSISS models.

2 Related Work

2.1 Remote Sensing Image Semantic Segmentation

Remote sensing image semantic segmentation (RSISS) is a core task in Earth observation analysis, with existing approaches primarily focusing on close-set segmentation using benchmark datasets like LoveDA [Wang 2021] and iSAID [Zamir 2019]. These datasets have enabled the development of deep learning-based models such as UNet and Vision Transformers [Yamazaki 2023; Zheng 2020]. To reduce annotation costs, partially supervised methods, including semi-supervised learning, few-shot learning, and domain adaptation, have been explored [??]. However, these methods remain limited to predefined class sets and lack the ability to generalize to unseen categories.

2.2 Open Vocabulary Semantic Segmentation

Open vocabulary semantic segmentation predicts pixel-wise labels for arbitrary categories specified at inference time, leveraging vision-language models like CLIP for strong semantic alignment. Existing methods include single-stage approaches that directly align dense visual features with text embeddings and two-stage approaches that decouple region proposal generation from semantic classification. While effective on natural images, these methods often degrade in performance on remote sensing imagery due to domain discrepancies.

2.3 Open Vocabulary RSISS and Datasets

The extension of open vocabulary segmentation to the remote sensing domain remains relatively underexplored. Existing remote sensing datasets are typically designed for close set segmentation and are often limited in scale or semantic diversity, making them unsuitable for evaluating open vocabulary settings. Although recent datasets such as FLAIR and SAMRS have attempted to expand the scope of RSISS, they either focus on coarse land cover categories or exhibit limited generalization across different segmentation tasks.

To address these limitations, the LandDiscover50K dataset was introduced as a large scale benchmark tailored for open vocabulary RSIS. LandDiscover50K includes 51,846 meticulously selected high resolution remote sensing images annotated across 40 object classes. The dataset is designed to overcome several limitations in existing RSIS datasets by addressing domain shifts through the incorporation of diverse sensors, resolutions, class variations, and ground sample distances. This diversity enables the dataset to capture substantial variability in spatial resolution, object scale, and scene composition, providing a challenging and comprehensive evaluation setting for open vocabulary segmentation models. Furthermore, LandDiscover50K enhances model robustness and generalization by integrating fine grained small target datasets and large scale land cover datasets, making it a valuable resource for advancing open vocabulary remote sensing image semantic segmentation.

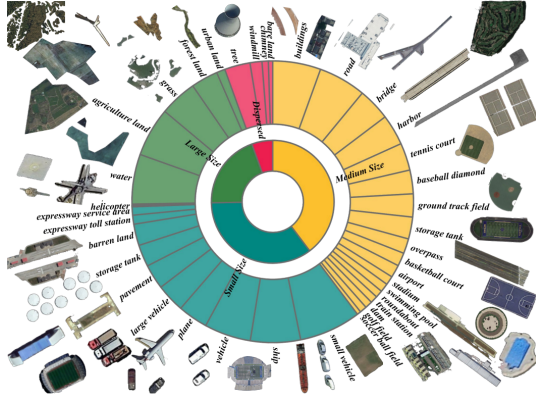


Figure 1: Illustration of the semantic class distribution and visual samples from LandDiscover50K. Sample images are strategically positioned adjacent to their corresponding semantic class tags for clarity.

3 Methodology

3.1 Overview of the Original Framework

The original framework proposed in “Towards Open-Vocabulary Remote Sensing Image Semantic Segmentation” [Zhang et al. 2023] addresses the challenge of segmenting remote sensing images under an open-vocabulary setting, where the model is required to recognize and segment semantic categories that are not explicitly seen during training. To tackle this problem, the authors introduce **GSNet**, a dual-stream segmentation framework that combines general semantic knowledge from vision–language models with domain specific representations tailored for remote sensing imagery.

GSNet Architecture. GSNet is built upon three main components: a *Dual-Stream Image Encoder (DSIE)*, a *Query-Guided Feature Fusion (QGFF)* module, and a *Residual Information Preservation Decoder (RIPD)*. The core idea is to jointly exploit the complementary strengths of generalist and specialist visual representations to achieve robust open-vocabulary segmentation.

Dual-Stream Image Encoder (DSIE). The DSIE consists of two parallel branches. The *generalist branch* leverages a pretrained

vision–language model, specifically CLIP with a ViT-based backbone, to extract high-level semantic features that are well aligned with textual class descriptions. This branch enables the model to generalize beyond a fixed set of categories by embedding visual features into a shared image text space.

In parallel, the *specialist branch* is designed to capture domain specific priors inherent to remote sensing imagery, such as spatial patterns, texture regularities, and object layouts. In the original framework, this specialist backbone is trained from scratch following the DINO self-supervised learning paradigm, using the LandDiscover50K dataset. By training the specialist encoder extensively on remote sensing data, the model acquires strong domain adaptive representations that complement the general semantic knowledge provided by the CLIP backbone.

Query-Guided Feature Fusion (QGFF). To effectively integrate the outputs of the generalist and specialist branches, GSNet introduces the QGFF module. This module uses text query embeddings derived from class names to guide the fusion process, allowing the model to selectively emphasize relevant visual features from both streams. Through query-guided attention mechanisms, QGFF aligns image features with semantic concepts expressed in natural language, enabling flexible segmentation of both seen and unseen categories.

Residual Information Preservation Decoder (RIPD). The fused features produced by QGFF are passed to the RIPD, which is responsible for generating dense pixel-level segmentation masks. The decoder preserves residual information from multiple feature levels to maintain spatial precision while refining semantic consistency. This design ensures that fine-grained details and high-level semantic cues are jointly retained in the final segmentation output.

Training Strategy in the Original Work. In the original GSNet framework, the training strategy reflects a clear separation of roles between the two backbones. The specialist branch is trained from scratch on the LandDiscover50K dataset prior to integration into GSNet, allowing it to serve as a strong domain-specific feature extractor. During GSNet training, the specialist backbone remains fixed, while the CLIP-based generalist backbone is lightly optimized to adapt semantic representations to the segmentation task. The remaining components, including the fusion and decoding modules, are trained from scratch. This training paradigm enables GSNet to balance semantic generalization and domain specialization, which is critical for open-vocabulary remote sensing image semantic segmentation.

3.2 Proposed Modifications

To enhance the GSNet framework for Open-Vocabulary Remote Sensing Image Semantic Segmentation (OVRSSS), we propose modifications to the generalist and specialist backbones while preserving the overall architecture and fusion strategy.

Generalist Backbone: For the generalist backbone, we consider various configurations of the CLIP model, including ViT-based architectures (e.g., ViT/B-16, ViT/L-14) and convolutional architectures (e.g., ResNet-50, ResNet-101). These options provide flexibility in balancing semantic alignment and spatial precision. While ViT-based models excel in capturing global semantic relationships, ResNet-based models are well-suited for preserving fine-grained

spatial details. In our implementation, the CLIP backbone is partial fine-tuned during training, with a learning rate of 2×10^{-6} , to adapt its generalist features to the remote sensing domain while retaining its open-vocabulary capability.

Specialist Backbone: For the specialist backbone, we propose two main approaches to leverage the strengths of DINOv3 for open-vocabulary remote sensing image semantic segmentation. The first approach involves training the model from scratch on the LandDiscover50K dataset, following the methodology outlined in the original GSNet framework. This strategy would allow the model to fully adapt to the unique characteristics of the dataset, but it is computationally intensive and requires significant resources.

The second approach considers using pretrained DINOv3 models specifically tailored for satellite imagery, such as those trained on the SAT-493M dataset. Within this category, two versions of the model could be evaluated: one with approximately 300 million parameters and another with 6.7 billion parameters. These pretrained models offer distinct advantages, with the larger model potentially providing greater capacity for capturing complex semantic and spatial relationships, while the smaller model balances performance with computational efficiency.

By narrowing the focus to these two strategies, we aim to provide a clear and practical framework for selecting and adapting the specialist backbone. These proposed approaches align with the overall goals of the GSNet framework while offering flexibility for future exploration and optimization.

3.3 Chosen Approaches

In this section, we outline the specific choices made for the generalist and specialist backbones in our implementation of the GSNet framework. These choices were guided by the unique requirements of open-vocabulary remote sensing image semantic segmentation and the strengths of the selected models.

For the generalist backbone, we selected the ResNet-101 architecture from CLIP. This choice was motivated by its compatibility with the output dimension size required for seamless integration with the specialist backbone and the fusion modules. Additionally, the convolutional nature of ResNet-101 is well-suited for capturing fine-grained spatial details in high-resolution remote sensing images. Unlike the ViT-based architectures commonly used in CLIP, ResNet-101 provides a balance between semantic alignment and spatial precision, making it a strong candidate for the generalist branch. The ResNet-101 backbone is partially fine-tuned during training, with a learning rate of 2×10^{-6} , to adapt its generalist features to the remote sensing domain while retaining its open-vocabulary capability.

For the specialist backbone, we chose the ViT-L/16 architecture from DINOv3, specifically the distilled model pretrained on the SAT-493M satellite-image dataset. This model was selected for its advanced self-supervised training pipeline, which is tailored for geospatial and Earth observation tasks. The ViT-L/16 backbone outputs dense patch-level features with a resolution of H/16, W/16, aligning well with the generalist branch and facilitating effective feature fusion. With approximately 300 million parameters, this backbone offers a robust yet manageable solution for extracting domain-specific features from remote sensing imagery. The decision

to use the 300M-parameter model over the larger 6.7B-parameter model was driven by the limited computational resources available, making the smaller model a more practical choice while still meeting the requirements of the task.

A critical decision was made regarding the adaptation of the DINOv3 specialist backbone to the LandDiscover50K dataset. While using the model as-is in a zero-shot setting was initially considered, it became evident that this approach would not fully leverage the dataset's unique characteristics. Instead, we determined that **fine-tuning** the DINOv3 model is a more effective strategy to adapt its pretrained features to the specific requirements of the task. However, fully fine-tuning all layers of the model is computationally expensive, and we risk changing the pretrained model weights such that it forgets previously learned representations. Therefore, we adopted a partially fine-tuning approach, which balances computational efficiency with the need for domain-specific adaptation. The extent of fine-tuning and its impact on performance will be analyzed and reported in subsequent sections.

These choices reflect a deliberate effort to integrate generalist and specialist representations that complement each other, leveraging the strengths of both convolutional and transformer-based architectures to address the challenges of open-vocabulary remote sensing image semantic segmentation.

3.4 Implementation Details

The Query-Guided Feature Fusion (QGFF) module and Residual Information Preservation Decoder (RIPD) remain unchanged. To align features, the ResNet-101 generalist backbone outputs 512-dimensional feature maps, while the DINOv3 specialist backbone outputs 1024-dimensional patch embeddings, which are projected to 512 dimensions for seamless fusion. This alignment preserves the integrity of the original GSNet design while supporting alternative backbone configurations.

4 Experiments

4.1 Dataset and Evaluation Protocol

In this study, the LandDiscover50K dataset is utilized exclusively for training purposes. This dataset provides a robust foundation for open-vocabulary remote sensing image semantic segmentation, with its diverse and meticulously annotated high-resolution images spanning 40 object classes. It ensures the model is exposed to a wide range of semantic and spatial variations during training.

For testing, we evaluate the model's performance on four representative datasets, as highlighted in the original paper, to simulate real-world scenarios and assess cross-dataset generalization:

- **FLAIR (Garioud et al. 2024):** A large-scale dataset covering diverse land cover types, enabling the evaluation of generalization to broad semantic categories.
- **FAST (Wang et al. 2024):** Specializes in fine-grained object segmentation, providing insights into the model's ability to handle detailed and intricate segmentation tasks.
- **ISPRS Potsdam (ISPRS 2013a):** Focuses on in-vocabulary performance with high category similarity to the training set, offering a benchmark for evaluating segmentation accuracy on familiar classes.

- **FloodNet (Rahnemoonfar et al. 2021):** A dataset dedicated to post-flood analysis, testing the model’s adaptability to domain-specific challenges.

To better understand the relationship between the training dataset (LandDiscover50K) and the validation datasets, we computed the label-set similarity scores. These scores quantify the overlap between the semantic categories in LandDiscover50K and those in the validation datasets, providing insights into the generalization capabilities of the model. Table 1 summarizes the similarity scores.

Table 1: The label-set similarity between LandDiscover50K and validation datasets.

Dataset	Label-set Similarity
FAST	0.79
FLAIR	0.69
Potsdam	0.69
FloodNet	0.71

To evaluate the performance of the GSNet framework and its extensions, we use the **Mean Intersection over Union (mIoU)** metric. This metric calculates the average IoU across all classes, providing a comprehensive measure of segmentation accuracy. It is particularly useful for assessing performance on both seen and unseen classes. In the open-vocabulary setting, special attention is given to the model’s performance on unseen classes. The mIoU metric is computed separately for seen and unseen categories to analyze the generalization capability of the framework. This distinction allows us to assess how effectively the model leverages generalist and specialist representations to segment novel semantic categories.

4.2 Training Methodology

The training process for the GSNet framework is described in terms of iterations rather than epochs. An iteration corresponds to a single training step where the model processes a batch of images, computes predictions, calculates the loss, and updates weights via backpropagation. For this study:

- **Batch size:** 2 images per iteration.
- **Total iterations:** 30,002, resulting in 60,004 total image views.
- **Epochs:** Approximately 1.16 epochs, calculated as total image views divided by the dataset size (51,846 images). Increasing the batch size to 4 would result in 120,008 total image views, extending the training length to approximately 2.31 epochs. This iteration-based approach ensures consistency with the original GSNet implementation and allows for efficient utilization of computational resources.

4.3 Experimental Setup

To evaluate the replication and extension of the GSNet framework, we conducted experiments using the LandDiscover50K dataset exclusively for training. This dataset provides a diverse benchmark for open-vocabulary remote sensing image semantic segmentation, with over 50,000 high-resolution images annotated across a wide range of semantic categories. The entire dataset was used for

training, relying on pretrained hyperparameters without additional validation tuning.

The GSNet framework was trained for 30K iterations, following the default setting of the original implementation. Training iterations refer to the number of optimization steps rather than epochs, ensuring a fair comparison with the original work. By adhering to this fixed schedule, we isolated performance differences arising from backbone replacements and fine-tuning strategies, rather than training duration.

In the original GSNet framework, the specialist model was trained from scratch for 300 epochs on the LandDiscover50K dataset, fully adapting to its unique characteristics but requiring significant computational resources. Our approach focuses on leveraging pre-trained models to achieve comparable results with reduced training time and resource requirements.

Baseline Configuration: In the original paper, the authors evaluated nine configurations of the GSNet framework by varying the training modes (full, attention, and frozen) for both the generalist backbone (CLIP ViT/B-16) and the specialist backbone (DINO ViT-B/8). Among these, the best-performing configuration was identified as the generalist backbone trained with attention mechanisms to enhance semantic alignment, while the specialist backbone was kept frozen to leverage its pretrained domain-specific features without additional fine-tuning.

In our work, we adopted the same approach for the generalist backbone, training the CLIP ViT/B-16 with attention mechanisms. For the specialist backbone, we extended the study by replacing the DINO ViT-B/8 backbone with DINO v3 and systematically exploring its three training configurations: full, attention, and frozen. Unlike the original work, which evaluated all nine combinations of training modes for both the generalist and specialist backbones, we focused on a fixed configuration for the generalist backbone (attention) and varied only the training modes for the specialist backbone. This approach allowed us to analyze the impact of DINO v3 configurations while maintaining consistency in the generalist backbone.

DINOv3

Fine-Tuning: Initially, the DINOv3 model pretrained on satellite images was used as-is in a zero-shot setting. However, it provided worse results compared to the DINOv1 model, highlighting the need for adaptation. To address this, we fine-tuned the DINOv3 model by unfreezing only the last two layers and training it on the LandDiscover50K dataset. This selective fine-tuning approach allowed us to adapt the pretrained features to the specific characteristics of the dataset while minimizing the computational overhead. The fine-tuning process focused on enhancing the model’s ability to capture domain-specific patterns while retaining its generalization capabilities. The fine-tuning was performed as a supervised task by training a segmentation head on top of the two fine-tuned layers, enabling the model to learn segmentation-specific features effectively.

Evaluation:

The evaluation of the fine-tuned DINOv3 backbone and its integration into the GSNet framework is conducted in two stages:

1. **Fine-Tuning Evaluation:** During the fine-tuning process, the model’s performance is monitored using the CrossEntropyLoss

function. The average training loss per epoch is recorded to assess the adaptation of the DINOv3 backbone to the LandDiscover50K dataset. This evaluation ensures that the fine-tuned model effectively captures domain-specific patterns while retaining its generalization capabilities. Additionally, validation metrics, if available, can provide further insights into the model's performance during fine-tuning.

2. GSNet Evaluation: The fine-tuned DINOv3 backbone is integrated into the GSNet framework, and the overall performance is evaluated using the Mean Intersection over Union (mIoU) metric. mIoU is computed across all classes to measure the overlap between predicted and ground truth segmentation masks. Special attention is given to the model's performance on unseen classes, reflecting its open-vocabulary segmentation capability. The evaluation is conducted on external datasets, including FLAIR, FAST, ISPRS Potsdam, and FloodNet, to assess the generalization of the framework.

CLIP CNN

GSNet Training: The CLIP CNN backbone was integrated into the GSNet framework to evaluate its performance as a generalist backbone. The training process focused on optimizing the QGFF module and RIPD while leveraging the pretrained CLIP CNN backbone for feature extraction. Additionally, the CLIP backbone was lightly trained during GSNet training with a learning rate of 2×10^{-6} , as mentioned in the methodology section. This configuration aimed to assess the balance between semantic alignment and spatial precision provided by the convolutional architecture.

Evaluation: The evaluation of the GSNet framework with the CLIP CNN backbone focuses on its performance in open-vocabulary segmentation tasks. The Mean Intersection over Union (mIoU) metric is used to measure the overlap between predicted and ground truth segmentation masks, with special attention given to unseen classes. The evaluation is conducted on external datasets, including FLAIR, FAST, ISPRS Potsdam, and FloodNet, to assess the generalization capability of the framework. The results are compared with other backbone configurations to highlight the strengths and limitations of using the CLIP CNN backbone without fine-tuning.

Experimental Environment: All experiments were conducted using the **PyTorch framework** on **NVIDIA GPUs**. Evaluation metrics included **mean Intersection over Union (mIoU)** and **pixel accuracy**, focusing on the model's ability to generalize to unseen semantic categories using external testing datasets. The computational setup ensured consistent and reproducible results across all experimental configurations.

4.4 Comparative Experiments

To evaluate the effectiveness of the proposed modifications, we conducted comparative experiments between the baseline GSNet configuration and the modified models. The experiments were designed to assess the impact of replacing the generalist and specialist backbones on segmentation accuracy, particularly for unseen classes.

Baseline Configuration: The baseline GSNet framework utilized the CLIP ViT/B-16 as the generalist backbone and the DINOv1 ViT-B/8 as the specialist backbone. This configuration served as the reference point for evaluating the proposed modifications.

Modified Configurations: The modified configurations included replacing the generalist backbone with ResNet-101 and the specialist backbone with DINOv3 variants. Each configuration was evaluated under consistent training protocols, datasets, and evaluation metrics to ensure a fair comparison.

Evaluation Metrics: The performance of each configuration was assessed using the Mean Intersection over Union (mIoU) metric, with a focus on both seen and unseen classes.

4.5 Exploring CLIP ViT L/14 with DINOv3

In addition to the configurations explored in the original GSNet framework, we extended our investigation by integrating a larger variant of the CLIP model, CLIP ViT L/14, as the generalist backbone. This configuration was paired with the fine-tuned DINOv3 specialist backbone, which was adapted to the dataset for 15 epochs as described earlier. The goal was to evaluate the potential of CLIP ViT L/14 for enhancing open-vocabulary remote sensing image semantic segmentation.

The training process for this configuration followed the same protocol as the other experiments, with the GSNet framework trained for 30K iterations. The CLIP ViT L/14 backbone was trained in three configurations: "Attention-Attention," "Freeze-Freeze," and "Attention-Freeze," where the first term refers to the training mode of the generalist backbone and the second term refers to the training mode of the specialist backbone. This ensured consistency in the experimental setup and allowed for a fair comparison of the results.

By leveraging the larger architecture of CLIP ViT L/14, we aimed to investigate its ability to capture more nuanced semantic and spatial relationships, which are critical for effective segmentation in open-vocabulary settings. Preliminary results indicate that this configuration exhibits mixed performance, with improvements observed in certain datasets while others showed no significant gains. These findings highlight the importance of balancing model complexity with task-specific requirements and suggest that further optimization may be needed to fully exploit the potential of CLIP ViT L/14 in the GSNet framework.

5 Quantitative Results

Fine-Tuning

Table 2 presents the mIoU results of GSNet trained for 5K iterations, with DINOv3 fine-tuned for varying numbers of epochs. Both CLIP and DINOv3 were set to the Attention mode for this experiment.

Table 2: mIoU (%) of GSNet trained for 5K iterations with varying DINOv3 fine-tuning epochs on the FloodNet dataset.

CLIP Mode	DINOv3 Fine-Tuning Epochs	mIoU (%)
Attention	0	28.401
Attention	5	29.012
Attention	10	31.059
Attention	15	34.124

The results in Table 2 demonstrate the impact of fine-tuning DINOv3 on GSNet performance under a reduced training budget. With

no fine-tuning, the model achieves an mIoU of 28.401%, indicating that pretrained DINOv3 features provide a strong baseline.

As the number of fine-tuning epochs increases, performance improves consistently, reaching 34.124% after 15 epochs. This highlights the importance of adapting the specialist backbone to the target dataset, even under resource-constrained training conditions. The results further illustrate the trade-off between computational cost and accuracy, showing that modest fine-tuning can yield substantial gains when training iterations are limited.

DINOv3

Table 3 reports the mIoU results of GSNet with different DINOv3 training modes. CLIP is fixed to the Attention configuration, which was shown to be optimal in the original GSNet work.

The Freeze mode underperforms the benchmark average, indicating that frozen pretrained features alone are insufficient for optimal performance. Partial fine-tuning in the Attention mode reduces this gap, while Full fine-tuning slightly exceeds the benchmark average.

Overall, the results indicate limited sensitivity to the DINOv3 training mode, with all configurations achieving similar average mIoU around 27%. Full fine-tuning provides only marginal gains over selective or frozen configurations, highlighting GSNet's stability and the efficiency of partial fine-tuning. However, all DINOv3-based configurations remain below the original GSNet all-time benchmark average, indicating that scaling pretrained specialists alone does not fully recover the performance achieved by a domain-specific specialist trained from scratch.

CLIP CNN

In this ablation study, we evaluated the performance of the GSNet framework using CLIP CNN as the generalist backbone and DINOv1 (trained on LandDiscover50K) as the specialist backbone. The primary objective was to analyze the impact of replacing the generalist backbone with a convolutional architecture for open-vocabulary remote sensing image segmentation. Unlike previous experiments, DINOv1 was frozen during training, and the CLIP CNN backbone was trained in the "Attention" mode.

The GSNet framework was trained for 30K iterations, consistent with the default setting of the original GSNet implementation. This ensures a fair comparison with other experiments and isolates the effect of the CLIP CNN backbone on segmentation performance.

The results, presented in Table 4, indicate that the performance of GSNet with CLIP CNN as the generalist backbone is significantly lower compared to the benchmark average. The average mIoU achieved across the four datasets is 4.58%, which is 26.67% below the benchmark average. This substantial performance gap highlights the limitations of using a convolutional architecture as the generalist backbone in the GSNet framework, particularly when the specialist backbone is frozen and no additional fine-tuning is applied.

Furthermore, the results demonstrate that CLIP CNN disastrously underperformed across all datasets. This underperformance can be attributed to the convolutional architecture's inability to effectively capture the complex spatial relationships required for open-vocabulary segmentation tasks. Addressing this limitation would require significant modifications to the fusion module to better integrate the generalist and specialist backbones. However, such modifications fall outside the scope of this project.

CLIP ViT L/14 with DINOv3

To complement the experiments conducted with CLIP ViT B/16, we evaluated the GSNet framework using the larger CLIP ViT L/14 as the generalist backbone, paired with a fine-tuned DINOv3 specialist backbone. The objective was to assess whether scaling the generalist backbone further improves open-vocabulary remote sensing segmentation performance. The GSNet framework was trained for 30K iterations, and DINOv3 was fine-tuned for 15 epochs, following the same protocol as previous experiments.

Table 5 reports the mIoU results for three configurations: Attention–Freeze, Freeze–Freeze, and Attention–Freeze. Among them, the *Attention–Freeze* configuration achieves the strongest overall performance observed in this study. Although its average mIoU of 27.85% remains below the benchmark average due to reduced performance on ISPRS Potsdam, it surpasses the benchmark individually on **FloodNet**, **FAST**, and **FLAIR**, demonstrating consistent gains across multiple open-vocabulary evaluation datasets.

The *Attention–Attention* configuration performs slightly worse, with an average mIoU of 26.02%, indicating that jointly fine-tuning both backbones does not yield additional benefits at this scale.

Finally, the Freeze–Freeze configuration, despite underperforming in terms of average mIoU, achieves higher scores than the Attention–Freeze setting on **FLAIR**, **FAST**, and **ISPRS Potsdam**. This indicates that freezing both backbones can preserve spatial detail and dataset-specific biases that benefit certain benchmarks. However, this configuration suffers a substantial performance drop on FloodNet, which largely accounts for its lower overall average. Moreover, despite relative improvements across configurations, performance on ISPRS Potsdam remains significantly below that reported in the original GSNet work, underscoring the persistent challenge of ultra-high-resolution segmentation when scaling pre-trained backbones.

Overall, these results indicate that scaling the generalist backbone to CLIP ViT L/14, when combined with selective fine-tuning of the DINOv3 specialist, yields the most competitive open-vocabulary segmentation performance in this work. The remaining performance gap on ISPRS Potsdam further reinforces the trade-off between semantic generalization and spatial precision observed throughout our experiments.

6 Conclusion

This work revisited open-vocabulary remote sensing image semantic segmentation by replicating and extending the GSNet framework with scaled pretrained backbones. Specifically, we replaced the specialist DINOv1 model trained from scratch with larger pretrained DINOv3 variant and explored selective fine-tuning strategies, while also scaling the generalist backbone from CLIP ViT-B/16 to ViT-L/14. Our goal was to assess whether reusing and scaling foundation models can improve generalization while reducing training cost.

Across experiments, GSNet exhibited strong stability with respect to specialist training configurations, with average mIoU remaining around 27% under a fixed training budget. Selective fine-tuning of DINOv3 consistently provided a favorable trade-off between performance and efficiency, while full fine-tuning yielded only marginal gains. Importantly, all DINOv3-based configurations remained below the original GSNet all-time benchmark average,

Table 3: mIoU (%) of GSNet with different DINOv3 training modes compared to the benchmark average. CLIP is fixed to Attention.

CLIP	DINOv3 Mode	FloodNet	FAST	Potsdam	FLAIR	Average	Benchmark Avg	Difference
Attention	Freeze	38.2	15.4083	37.2095	19.5551	27.5932	31.25	-3.6568
Attention	Attention	36.8173	15.0792	35.4761	22.9478	27.5801	28.04	-0.4599
Attention	Full	38.5336	15.4324	36.6977	17.6921	27.089	27.04	0.049

Table 4: mIoU (%) of GSNet with CLIP CNN as the generalist backbone and DINOv1 as the specialist backbone. DINOv1 is frozen, and CLIP CNN is trained in the "Attention" mode.

CLIP Mode	FloodNet	FAST	Potsdam	FLAIR	Average	Benchmark Avg	Difference
Attention	3.2935	2.4179	10.3902	2.2193	4.5802	31.25	-26.6698

Table 5: mIoU (%) of GSNet with CLIP ViT L/14 as the generalist backbone and fine-tuned DINOv3 as the specialist backbone.

Configuration	FloodNet	FAST	Potsdam	FLAIR	Average	Benchmark Avg	Difference
Attention-Freeze	43.4789	17.2405	30.159	20.5263	27.8512	31.25	-3.3988
Attention-Attention	41.6734	17.7978	26.6669	17.9295	26.0169	28.04	-2.0231
Freeze-Freeze	28.1215	18.0169	31.6137	21.5861	24.8346	22.1	2.7346

indicating that scaling pretrained specialists alone does not fully replace domain-specific training from scratch.

Scaling the generalist backbone to CLIP ViT L/14 led to the strongest open-vocabulary performance observed in this study. The Attention-Freeze configuration achieved consistent improvements over the benchmark on FAST, FLAIR, and FloodNet individually, representing the best-performing setting across most evaluation datasets. However, performance on ISPRS Potsdam remained substantially lower than that of the original GSNet, highlighting a persistent trade-off between semantic generalization and spatial precision when scaling pretrained models for ultra-high-resolution segmentation.

Overall, these results demonstrate that large pretrained foundation models, when combined with selective fine-tuning, offer a practical and efficient alternative to training specialist backbones from scratch for open-vocabulary remote sensing segmentation. At the same time, our findings expose clear limitations on ultra-high-resolution benchmarks, suggesting that future work should focus on improving spatial fidelity without sacrificing generalization.

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